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8-Puzzle Problem

The 8-puzzle has a 3×3 board with 8 numbered tiles and one blank. A tile next to the blank can be slid into it, which is the same as moving the blank up, down, left or right. The aim is to reach the goal configuration from the initial one.

1. Problem Formulation

States: Any arrangement of the 8 tiles and the blank on the 3×3 board (9! = 362,880 arrangements, of which exactly half, 181,440, are reachable from a given start).

Initial state: The starting configuration shown below.

Goal state: The target configuration shown below.

Operators: Move the blank Up, Down, Left or Right (only if it stays on the board).

Path cost: Each move costs 1, so the cost of a solution is the number of moves.

```
Initial      Goal
2 8 3        1 2 3
1 6 4        8 _ 4
7 _ 5        7 6 5
```

2. Solution (Breadth-First Search)

Breadth-first search expands the states level by level, so the first time it reaches the goal the path is the shortest. The solution found is 5 moves of the blank: **Up, Up, Left, Down, Right**.

```
Start        1.Up         2.Up
2 8 3        2 8 3        2 _ 3
1 6 4        1 _ 4        1 8 4
7 _ 5        7 6 5        7 6 5

3.Left       4.Down       5.Right
_ 2 3        1 2 3        1 2 3
1 8 4        _ 8 4        8 _ 4
7 6 5        7 6 5        7 6 5
```

(_ is the blank. The label above each board is the move number and the direction the blank moved.)

3. Result

The goal is reached in **5 moves**. Since breadth-first search is complete and optimal when every move costs 1, no shorter solution exists. It expanded 34 states before reaching the goal.